

PlantPulse

Compound-risk intelligence for industrial safety

A gas leak masked by ventilation failure. A bearing failure on an overdue-maintenance compressor. An overpressure risk while the relief path is degraded.

Every one built so no single sensor ever crosses its own alarm threshold — and PlantPulse still catches all three.

100% detected · **99.8 min** median lead time

ET AI Hackathon 2.0 — Problem Statement #1, Industrial Safety Intelligence

The insight

Post-mortems on real industrial incidents keep saying the same thing:
"every individual system worked as designed."

- The permit system didn't know the compressor was oscillating.
- The gas detector didn't know a worker had just entered the zone.
- The maintenance scheduler didn't know a relief valve was already offline.

Individually normal. Combined, lethal. A conventional single-sensor alarm **cannot see this** — it only ever has one signal to look at.

Three real compound archetypes

Each validated at generation time: every affected channel stays under its own single-channel alarm threshold for the whole incident window, or the code refuses to run.

Archetype	Compounding factors
Slow gas leak, masked ventilation	Gas trending up + wind near-calm + unpermitted worker entry
Early bearing failure	Vibration + temp trending up + maintenance overdue 6 days
Overpressure, degraded relief	Pressure + temp trending up + relief valve mid-test + active hot-work permit

How it works

```
Python engine (offline, seeded) --exports--> JSON --> Next.js control room
  6 agents + IsolationForest                                plays precomputed demo
  fused into one score                                       live what-if slider
      |                                                         |
      +----- FastAPI (upside only) -----+
```

- **6 explainable agents** (sensor, equipment, permit, maintenance, worker, weather)
- **1 genuinely multivariate model** (IsolationForest, trained only on 30 quiet days — never on the incidents)
- **Fused score anchored at a calibrated alert threshold** — not a raw percentile, which is uniform on normal data by construction

The control room

Live risk gauge, plant twin, geospatial heatmap, and ranked contributor bars — all driven by the same exported JSON, deterministically.

(Live/Demo/Knowledge tabs — see recorded demo)

Zone turns **RESTRICTED** the moment compound risk crosses threshold.
Workers render as dots; an unpermitted entry renders in red.

Explainable by construction

A real knowledge graph (Worker → Zone → Permit , Asset → Channel , Asset → Maintenance Task) built fresh from plant state every run — not a static diagram.

"Worker 05 enters Tank Farm (Z1) — Z1's standing entry permit P-01 is held by Worker 01, not Worker 05; TANK-01 shows Gas Concentration rising; wind has dropped to near-calm, stalling dispersion."

Generated from real graph traversal + real event data. Every incident.

The metric that matters

20 seeded evaluation days, 60 archetype instances, 20 held-out quiet days for false-alarm rate. Source: `engine/metrics/results.json` .

Method	Detected	Median Lead Time	False Alarms/Day
PlantPulse (hero)	100%	99.8 min	7.05
Threshold baseline	0%	never fires	0.00
Rolling z-score baseline	100%	92.4 min	8.30

Wins on lead time on **2 of 3 archetypes individually** — the third (fastest-onset overpressure) trails by ~1 minute, reported honestly.

From alert to action

Ask "Why is Tank Farm unsafe right now?" →

Cause (from the knowledge graph) → **Recommendation** (top SOP steps)

→ **SOP-GAS-01**, cited to OISD-STD-105 & Factories Act 1948 §36 →

Estimated risk reduction: computed by rescoring a real mitigated counterfactual through the same model, not invented → **One-click PDF incident report**.

Runs fully offline from a pre-generated cache.

Business impact — labeled honestly

(All figures: simulation output — measured count × disclosed assumption. Full basis in `data/cost_basis.md`.)

Figure	Value	Basis
Near-misses prevented/yr	12	3 archetypes × 4/yr assumed × 100% detected
Downtime avoided/yr	48h	4h assumed per incident
Cost avoided/yr	₹36,00,000	₹3,00,000 assumed per incident
Unsafe entries flagged	20/20	Measured, not assumed

No "lives saved" figure — a life-safety claim tied to a simulation would be dishonest regardless of labeling.

The ask

PlantPulse is a fusion layer, not a new sensor: it recalibrates against a plant's own historian and its own asset/zone topology — no new hardware.

We're asking for the chance to pilot this against one real facility's historian data.

Repo: github.com/sahaj645/et-hack